

Generation Interconnection System Impact Study Report

For Queue Project Q20-1384

Sugar Island 100 kV

July 30, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1384. The Interconnection Customer proposes a 153.1 MW battery energy storage facility interconnecting at the Sugar Island 100 kV substation in the Ohio area. The request was received on January 13, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 431 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 30 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$132,841,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Vitulli – Anaverde 100 kV	Canal Rd – Newport Tie 230 kV	173.7	141	14.3
Himes – Anaverde 100 kV	Canal Rd – Newport Tie 230 kV	168.4	138	14.3
Nampa – Himes 100 kV line	Canal Rd – Newport Tie 230 kV	151.4	128	14.3
Sugar Island – Harem 100 kV	Newport Tie – Canal Road 230 kV	150.5	102	14.3
Madisonburg – Newport Tie Substation 100 kV Line 1	Newport Tie – Madisonburg 100 kV	149.1	161	7.3
Newport Tie – Madisonburg 100 kV Cir. 2	Newport Tie – Madisonburg 100 kV Ckt 1	149.1	161	7.3
Newport Tie – Madisonburg 100 kV #3	Newport Tie – Madisonburg 100 kV Line 1	149.1	161	7.3
Nampa – Harlem Substation 100 kV	Canal Road – Newport Tie 230 kV Line	146.3	118	14.3
Suga Island – Nimisila Substation 100 kV L	Vitulli – Anaverde 100 kV	139.8	136	31.6
Rosedle – Sugar Island 100 kV Line	Wayview – Sugar Island 100 kV	132.3	145	49.4
Bell Rock Rd – Moore 100 kV	Portage Lakes 100-1 kV Auto	131.6	142	26.5
Newport Tie 230/100 kV XFMR	Newport Tie 230/100 kV XFMR	120.4	229	6.3
Newport Tie 230/100kV Transformer Bank #2	Newport Tie 230 kV/100 kV Auto Bank 1	120.4	229	6.3
Newport Tie 230/100kV Auto #3	Newport Tie 230 kV/100 kV Autotransformer #1	120.4	229	6.3
Wayview – Sugar Island 100 kV	Rosedale – Sugar Island 100 kV	119.3	119	38.9
Portage Lakes – Nimisila 100 kV line	New Moccasin Cr – Portage Lakes 100 kV	118.0	136	24.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wilkins Corner – Rittman 100 kV	Portage Lakes 100-1 kV Transformer	113.5	131	8.8
Bain – Rittman 100 kV	Portage Lakes 100 kV/1 kV Transformer	112.0	133	8.8
Madisonburg – Wilkins Corner 100 kV line	Portage Lakes 100/1kV Auto	111.8	144	8.8
Green – Wayview 100 kV	Rosedale – Sugar Island 100 kV L	109.1	106	38.9
Portage Lakes Substation 100 kV/1 kV Auto	Madisonburg – Wilkins Corner 100 kV	108.7	228	24.3
Weedpatch – Portage Lakes 230 kV	Point Arena – W Canton 500 kV	107.6	875	8.9
Water Substation – S Akron 100 kV	S Akron – Water 100 kV line	105.7	157	13.4
Moore – South Akron 100 kV Circuit 2	S Akron – Moore 100 kV Cir. 1	105.1	154	14.2
Knox – Aes 230 kV	Atascocita – N Cambridge 230 kV	103.9	468	5.6
New Moccasin Cr – Portage Lakes 100 kV	Portage Lakes – Nimisila 100 kV	103.8	131	19.9
Greentown Substation – Bell Rock Rd 100 kV	Rosedale – Sugar Island 100 kV	103.2	130	15.0
Bain – Forward Energy Ctr 100 kV Line	Portage Lakes 100/1kV XFMR	103.1	125	8.8
Moore – South Akron Substation 100 kV Cir. 1	Moore – South Akron 100 kV #2	102.1	159	14.2
S Akron – Water 100 kV Ckt 1	South Akron Substation – Water 100 kV Cir. 2	101.6	164	13.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Vitulli – Anaverde 100 kV	Rebuild line with higher-capacity conductor	\$1,824,000	100% (\$1,824,000)
Himes – Anaverde 100 kV	Rebuild line with higher-capacity conductor	\$2,690,000	100% (\$2,690,000)
Nampa – Himes 100 kV line	Rebuild line with higher-capacity conductor	\$4,660,000	100% (\$4,660,000)
Sugar Island – Harem 100 kV	Rebuild line with higher-capacity conductor	\$2,526,000	100% (\$2,526,000)
Newport Tie – Madisonburg 100 kV line	Rebuild line with higher-capacity conductor	\$5,971,000	100% (\$5,971,000)
Newport Tie – Madisonburg 100 kV Cir. 2	Rebuild line with higher-capacity conductor	\$5,219,000	100% (\$5,219,000)
Newport Tie – Madisonburg 100 kV L	Rebuild line with higher-capacity conductor	\$13,381,000	100% (\$13,381,000)
Nampa – Harlem Substation 100 kV	Rebuild line with higher-capacity conductor	\$4,634,000	100% (\$4,634,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sugar Sland – Nimisila 100 kV	Rebuild line with higher-capacity conductor	\$2,531,000	93% (\$2,362,000)
Rosedale – Sugar Island 100 kV	Rebuild line with higher-capacity conductor	\$7,271,000	73% (\$5,296,000)
Moore Substation – Bell Rck Rd Substation 100 kV	Rebuild line with higher-capacity conductor	\$3,396,000	54% (\$1,817,000)
Newport Tie 230/100 kV Autotransformer #1	Replace transformer with higher-rated unit	\$11,322,000	100% (\$11,322,000)
Newport Tie 230/100kV Transformer Bank #2	Replace transformer with higher-rated unit	\$15,696,000	100% (\$15,696,000)
Newport Tie 230/100kV Auto #3	Replace transformer with higher-rated unit	\$8,385,000	100% (\$8,385,000)
Wayview – Sugar Island 100 kV	Rebuild line with higher-capacity conductor	\$2,051,000	100% (\$2,051,000)
Nimisila – Portage Lakes 100 kV	Rebuild line with higher-capacity conductor	\$8,414,000	92% (\$7,769,000)
Wilkins Corner – Rittman 100 kV	Rebuild line with higher-capacity conductor	\$10,497,000	100% (\$10,497,000)
Rittan – Bain 100 kV	Rebuild line with higher-capacity conductor	\$2,081,000	100% (\$2,081,000)
Madisonburg – Wilkins Corner 100 kV line	Rebuild line with higher-capacity conductor	\$7,740,000	100% (\$7,740,000)
Green – Wayview 100 kV	Reconductor line	\$923,000	100% (\$923,000)
Portage Lakes 100/1 kV Autotransformer	Replace transformer with higher-rated unit	\$4,189,000	92% (\$3,851,000)
Weedpatch – Portage Lakes 230 kV	Reconductor line	\$20,044,000	14% (\$2,740,000)
Water Substation – S Akron 100 kV	Reconductor line	\$1,436,000	87% (\$1,255,000)
Moore – S Akron Substation 100 kV	Reconductor line	\$963,000	87% (\$842,000)
Knox – Aes 230 kV	Reconductor line	\$5,733,000	45% (\$2,580,000)
New Moccasin Cr – Portage Lakes 100 kV	Reconductor line	\$1,489,000	100% (\$1,489,000)
Greentown Substation – Bell Rock Rd 100 kV	Reconductor line	\$594,000	35% (\$209,000)
Bain – Forward Energy Ctr 100 kV Line	Reconductor line	\$1,430,000	100% (\$1,430,000)
Moore – South Akron Substation 100 kV Cir. 1	Reconductor line	\$884,000	87% (\$773,000)
South Akron – Water 100 kV	Reconductor line	\$948,000	87% (\$828,000)
	Total allocated to project		\$132,841,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.